

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 1 – Prompt Engineering Challenge**

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Prompt Engineering Challenge
Weightage:	40% of CIA	Total Marks:	100 (2 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems. (BL1, BL2)</i>		
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Section / Batch:	2024- CSE(AI)	Date:	28-07-2026

Instructions to Students

- Answer 2 questions. Each question carries 50 Marks. Total: 100 Marks.
 - Write clear, well-structured prompts and explanations where asked.
 - Use your own original examples — do not copy from classmates or the internet verbatim.
 - Where a diagram or worked example is requested, label it clearly.
 - Illustrate prompting techniques (zero-shot, few-shot, chain-of-thought, role/system prompting, etc.) with concrete examples, not just definitions.
- Time allotted: 30 mins

Code of Conduct

I S.Sirivally / 192472371 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____ Sirivally.S ____

SECTION A – PROMPT ENGINEERING CHALLENGE

Q17. Comparative / Contrastive Evaluation Prompt

Write a prompt that gives the model two pieces of writing (or two code snippets) and asks it to evaluate both against specific criteria, then declare a winner with justification for each criterion.

```
>>>===== RESTART: C:/Users/ACER/RL-Lab/CO2-AT1_1.PY =====>>>Here are the comparisons for each criterion:>>>>>>1. Grammar**>>>Paragraph A: 2/10>>>Paragraph B: 10/10>>>Reason: Paragraph A lacks a complete sentence, whereas Paragraph B is grammatically correct with a clear subject-verb-object structure.>>>>>>2. Clarity**>>>Paragraph A: 2/10>>>Paragraph B: 9/10>>>Reason: Paragraph A is very vague, whereas Paragraph B clearly states the benefits of Artificial Intelligence in healthcare, making it easier to understand.>>>>>>3. Organization**>>>Paragraph A: 1/10>>>Paragraph B: 9/10>>>Reason: Paragraph A lacks organization, whereas Paragraph B is well-structured, with a clear topic sentence and supporting details.>>>>>>4. Vocabulary**>>>Paragraph A: 4/10>>>Paragraph B: 8/10>>>Reason: Paragraph A uses very general vocabulary, whereas Paragraph B uses more specific and technical terms like "disease diagnosis", "patient care", and "robotic surgery".>>>>>>5. Overall Quality**>>>Paragraph A: 2/10>>>Paragraph B: 9/10>>>Reason: Paragraph A is very weak in all aspects, whereas Paragraph B is well-written, clear, and informative.>>>>>>Conclusion**>>>Based on the comparisons, Paragraph B is significantly better than Paragraph A in all criteria. It provides clear and specific information about the benefits of Artificial Intelligence in healthcare, is well-organized, and uses more technical vocabulary. Therefore, I declare **Paragraph B** as the better paragraph with proper justification.>>>
```

```
from huggingface_hub import InferenceClient>>>client = InferenceClient(>>>    api_key="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">>>)>>>>>>prompt = "">>>Compare the following two paragraphs based on:>>>1. Grammar>>>2. Clarity>>>3. Organization>>>4. Vocabulary>>>5. Overall Quality>>>For each criterion:>>>- Give marks out of 10.>>>- Explain your reason.>>>Finally declare the better paragraph with proper justification.>>>>>>Paragraph A:>>>Artificial Intelligence is useful in healthcare.>>>>>>Paragraph B:>>>Artificial Intelligence improves disease diagnosis, patient care, robotic surgery, and drug c>>>"">>>>>>response = client.chat.completions.create(>>>    model="meta-llama/Llama-3.1-8B-Instruct",>>>    messages=[>>>        {"role": "user", "content": prompt}>>>    ],>>>    max_tokens=500>>>)>>>print(response.choices[0].message.content)>>>
```

Q20. Clarifying-Question Prompt (Ambiguity Resolution)

Write a prompt that, when given an underspecified request (e.g., "make this shorter"), instructs the model to ask a clarifying question first — instead of guessing at the user's intent — before attempting the task.

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32>>>Enter "help" below or click "Help" above for more information.>>>===== RESTART: C:/Users/ACER/RL-Lab/CO2-AT1_2.PY =====>>>What would you like to make shorter: a sentence, a document, a list, or something else?>>>>>>===== RESTART: C:/Users/ACER/RL-Lab/CO2-AT1_2.PY =====>>>What would you like to make shorter: a document, a sentence, or something else?>>>>>>===== RESTART: C:/Users/ACER/RL-Lab/CO2-AT1_2.PY =====>>>What would you like to make shorter?>>>|>>>
```

```
from huggingface_hub import InferenceClient>>>client = InferenceClient(>>>    api_key="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">>>)>>>>>>prompt = "">>>You are an AI assistant.>>>If the user's request is ambiguous or incomplete, do NOT guess.>>>Instead,>>>1. Ask exactly one clarifying question.>>>2. Wait for the user's response.>>>3. Only then perform the requested task.>>>>>>User:>>>Make this shorter.>>>"">>>>>>response = client.chat.completions.create(>>>    model="meta-llama/Llama-3.1-8B-Instruct",>>>    messages=[>>>        {"role": "user", "content": prompt}>>>    ],>>>    max_tokens=300>>>)>>>print(response.choices[0].message.content)>>>
```

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technique Selection	30%	Correctly applies zero-shot/one-shot/few-shot & CoT technique appropriate to each task	Applies techniques with minor mismatch to task	Limited understanding of which technique to use	Techniques applied incorrectly or not evident
Output Effectiveness	25%	Prompts produce accurate, task-appropriate outputs across all assigned tasks	Prompts work well for most tasks	Prompts work for only a few tasks	Prompts are largely ineffective
Prompt Craftsmanship	20%	Prompts are clear, specific, and reusable	Prompts are mostly clear	Prompts are vague or inconsistent	Prompts are poorly constructed
Prompt-Output Documentation	15%	Well-documented prompt-output pairs with clear rationale	Documented with minor gaps	Minimal documentation of prompts/outputs	No rationale provided for prompt choices
Design Rationale	10%	Clearly explains design choices and iterations made	Explains most design choices	Limited explanation of process	No explanation of process provided

Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness(25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale(10%)	100
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____